

Name:

FRANKLIN ONYEISE

TRANSPORT PHENOMENA I, CHE 311

QUIZ 2

CLOSED BOOK, TIME: 15 MINUTES

A Newtonian fluid is in laminar flow through a narrow slit formed by two vertical parallel walls a distance $2B$ apart. The wall at $x = -B$ is stationary, while the wall at $x = +B$ moves with a constant velocity v_B in z direction.

Make a shell momentum balance to obtain the velocity profile $v_z(x)$.

Model assumption:

- Steady state
- Neglect edge effects, assuming $B \ll W \ll L$
- Constant density ρ and viscosity μ
- $p(z=0) = p_0$, $p(z=L) = p_L$, and $\frac{\partial p}{\partial x} = \frac{\partial p}{\partial y} = 0$
- $v_z = v_z(x)$, and $v_x = v_y = 0$.

$$v_z(x) = \frac{1}{2\mu} \frac{dp}{dz} (x^2 - B^2) + \frac{v_B}{2B} (x+B)$$

$$p(z=0) = p_0 \quad p(z=L) = p_L$$

$$\frac{dp}{dz} = \frac{p_L - p_0}{L}$$

$$v_z(-B) = 0 \rightarrow \text{stationary wall}$$

$$v_z(B) = v_B \rightarrow \text{moving wall}$$

$$\tau_{xz} = \mu \frac{dv_z}{dx}$$

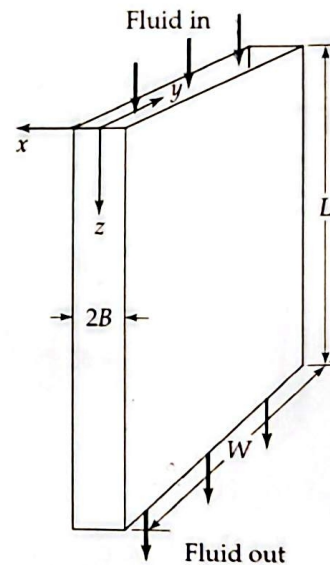
$$0 = -\frac{dp}{dz} + \frac{d}{dx} \left(\mu \frac{dv_z}{dx} \right)$$

$$0 = -\frac{dp}{dz} + \mu \frac{d^2 v_z}{dx^2}$$

$$\mu \frac{d^2 v_z}{dx^2} = \frac{dp}{dz}$$

$$\frac{d^2 v_z}{dx^2} = \frac{1}{\mu} \frac{dp}{dz}$$

$$\frac{dv_z}{dx} = \frac{1}{\mu} \frac{dp}{dz} x + C_1$$



60/100

Name:

$$V_2(x) = \frac{1}{2} p \frac{dp}{dz} x^2 + C_1 x + C_2$$

$$V_2(-B) = 0$$

$$0 = \frac{1}{2} p \frac{dp}{dz} B^2 - C_1 B + C_2$$

$$V_B \frac{1}{2} p \frac{dp}{dz} B^2 + C_1 B + C_2 \quad V_B = 2 C_1 B$$

$$C_1 = \frac{V_B}{2B}$$

$$C_2 = -\frac{1}{2} p \frac{dp}{dz} B^2 + \frac{V_B}{2}$$

$$V_2(x) = \frac{1}{2} p \frac{dp}{dz} (x^2 - B^2) + \frac{V_B}{2B} (x + B)$$